

AI-Powered Image Restoration for Edge and 5G Environments

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Declaration of Originality

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AI-Powered Image Restoration for Edge and 5G Environments

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Abstract

The expansion of fifth-generation (5G) communication networks and edge computing platforms has significantly increased the reliance on visual data for latency-sensitive applications such as network diagnostics, industrial monitoring, remote engineering analysis, and intelligent automation.

In operational environments, images are frequently captured and transmitted at low resolution due to device hardware limitations, power efficiency considerations, and bandwidth constraints. Although image restoration and super-resolution techniques can enhance the usefulness of such data, their deployment within real-world telecom networks raises challenges related to compute placement, latency, scalability, and operational governance.

This project investigates the deployment of AI-based image restoration within a telecom network context and evaluates different architectural choices for processing degraded visual data. Restoration at the device edge is examined and found to be constrained by hardware capability, ownership restrictions, and governance limitations, while centralized cloud-based processing introduces additional latency and bandwidth overhead. To address these limitations, a near-edge deployment strategy is explored, wherein image restoration is performed at operator-controlled edge or metro data centers.

Within this framework, both classical image restoration methods and deep learning-based approaches are studied, including convolutional neural networks (CNNs) and lightweight hybrid CNN-Transformer architectures designed for constrained compute environments. The techniques are evaluated based on reconstruction quality, robustness to noise and compression artifacts, and suitability for deployment in near-edge settings.

In addition, system-level considerations such as latency, scalability, and service ownership are analyzed. The study demonstrates that near-edge image restoration provides a practical compromise between performance, deployment feasibility, and operational efficiency for 5G-enabled visual applications.

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Acronyms, Abbreviations and Symbols

Acronyms and Abbreviations

5G	Fifth Generation Mobile Network
AI	Artificial Intelligence
AWGN	Additive White Gaussian Noise
BM3D	Block Matching and 3D Filtering
CNN	Convolutional Neural Network
DnCNN	Denoising Convolutional Neural Network
FFN	Feed-Forward Network
GPU	Graphics Processing Unit
HR	High Resolution
IoT	Internet of Things
LR	Low Resolution
MHSA	Multi-Head Self-Attention
MSE	Mean Squared Error
PSNR	Peak Signal-to-Noise Ratio
ReLU	Rectified Linear Unit
RSTB	Residual Swin Transformer Block
SSIM	Structural Similarity Index Measure
SRCNN	Super-Resolution Convolutional Neural Network
SwinIR	Swin Transformer for Image Restoration

Symbols

x	Clean (ground truth) image
y	Observed degraded/noisy image
$\hat{x}$	Restored (reconstructed) image
n	Additive noise
$\hat{n}$	Estimated noise
H	Degradation operator (e.g., blur kernel)
f_θ	Neural network function with parameters θ
σ	Standard deviation of noise
$\mathcal{N}(0, \sigma^2)$	Gaussian distribution with mean 0 and variance σ^2
$\mathcal{L}_{MSE}$	Mean Squared Error loss function
N	Number of pixels/samples
Q, K, V	Query, Key, Value matrices in attention mechanism
d_k	Dimension of key vectors

Chapter 1

Introduction

The convergence of edge computing and fifth-generation (5G) communication networks has significantly accelerated the generation, transmission, and analysis of visual data across distributed systems. Applications such as smart surveillance, industrial automation, remote diagnostics, network engineering, and intelligent transportation systems increasingly rely on images captured by resource-constrained devices, including cameras, mobile terminals, and Internet of Things (IoT) sensors.

To operate under constraints related to power consumption, storage capacity, and network bandwidth, these devices often transmit images at reduced resolution or with aggressive compression. Furthermore, visual data transmitted over wireless and edge networks is susceptible to various degradations, including noise, compression artifacts, motion blur, packet loss, and adverse environmental conditions. Such degradations negatively impact both human perception and the performance of downstream computer vision tasks, such as object detection, classification, anomaly detection, and diagnostic analysis.

Image restoration aims to recover high-quality and information-rich representations from degraded observations, thereby playing a critical role in enabling reliable visual intelligence in modern communication systems. Traditional restoration techniques, including spatial filtering, transform-domain denoising, and model-based optimization methods, rely on handcrafted priors and simplified assumptions about image structure and noise characteristics. While computationally efficient, these approaches often struggle to generalize to complex and diverse real-world degradation patterns.

Recent advances in deep learning have led to the development of data-driven restoration techniques based on convolutional neural networks (CNNs), encoder-decoder architectures, and attention mechanisms. These methods learn non-linear mappings from degraded inputs to high-quality reconstructions and have demonstrated significant performance improvements over classical approaches. However, deploying deep learning models in edge and 5G environments remains challenging due to strict latency constraints, limited computational and memory resources, and energy efficiency requirements.

Motivated by these challenges, this work investigates both classical and learning-based image restoration techniques, with a focus on identifying models suitable for deployment in edge-oriented and 5G-enabled systems. In particular, a hybrid CNN-Transformer architecture is

explored to leverage both local feature extraction and global contextual modeling while maintaining computational efficiency.

Although real-world edge and 5G environments introduce a wide range of degradation types—including compression artifacts, packet loss, motion blur, and transmission errors—this study focuses on a controlled subset of degradations to ensure systematic evaluation and reproducibility. Additive Gaussian noise is adopted as a baseline due to its mathematical tractability and widespread use in the literature, enabling standardized performance comparison across models.

To enhance practical relevance, a more realistic degradation model combining Poisson (shot noise) and Gaussian (read noise) is also considered. This hybrid noise model approximates sensor-level noise commonly observed in imaging devices, particularly in low-light and resource-constrained edge environments. While Gaussian noise alone does not fully capture transmission-related distortions such as packet loss or compression artifacts, it provides a controlled benchmark for analyzing model behavior under varying noise levels.

In practical 5G imaging pipelines, degradations may arise at multiple stages, including:

- **Image acquisition:** sensor-level noise introduced during capture
- **Compression:** artifacts arising from bandwidth-efficient encoding
- **Transmission:** packet loss and channel-induced distortions

The degradation models used in this work primarily approximate sensor-level noise while providing a controlled yet meaningful framework for evaluating restoration performance. This choice reflects a balance between experimental rigor and real-world applicability, enabling systematic analysis while retaining practical relevance.

Overall, the selected approach facilitates both controlled evaluation and meaningful insights for deployment in edge-based imaging systems, while also establishing a foundation for future work incorporating compression and transmission-level degradations.

Chapter 2

Literature Review

2.1 Introduction

Image restoration aims to recover a high-quality image from its degraded observation. In practical imaging systems, degradations may arise due to sensor noise, motion blur, defocus, compression artifacts, transmission losses, or adverse environmental conditions. In emerging 5G-enabled edge environments, restoration becomes increasingly critical, as images are often compressed or degraded to meet bandwidth and latency constraints.

This section reviews the evolution of image restoration techniques, beginning with classical signal processing approaches, followed by deep learning-based methods using convolutional neural networks (CNNs), and finally Transformer-based architectures. The strengths and limitations of each class of methods are analyzed to motivate the need for efficient hybrid models suitable for edge deployment.

2.2 Classical Image Restoration

2.2.1 Mathematical Formulation

Classical image restoration methods are grounded in signal processing and inverse problem theory. The degradation process is typically modeled as:

$$y = Hx + n \tag{2.1}$$

where y is the observed degraded image, x is the latent clean image, H represents the degradation operator (e.g., blur kernel), and n denotes additive noise.

2.2.2 Spatial Domain Methods

Early restoration approaches operate in the spatial domain and include mean filtering, median filtering, and adaptive filtering techniques. The Wiener filter improves denoising performance

by incorporating statistical assumptions about signal and noise distributions, providing a trade-off between noise suppression and detail preservation.

Blind deconvolution methods attempt to jointly estimate the blur kernel and the clean image. However, these approaches are highly sensitive to model inaccuracies and noise assumptions.

2.2.3 Frequency Domain Methods

Frequency-domain techniques leverage Fourier transforms to simplify convolution operations. While computationally efficient, they are sensitive to inaccuracies in the degradation model and tend to amplify noise in high-frequency components.

2.2.4 Sparse and Prior-Based Methods

Sparse representation and prior-based approaches introduce stronger regularization by leveraging assumptions about natural image statistics. These methods improve reconstruction quality but typically require computationally intensive optimization procedures.

2.2.5 Limitations of Classical Methods

Despite their strong theoretical foundation and interpretability, classical methods exhibit several limitations:

- Dependence on accurate degradation models
- Sensitivity to noise assumptions
- Limited ability to handle complex real-world degradations
- Over-smoothing of fine textures

These limitations restrict their applicability in dynamic and resource-constrained 5G edge environments.

2.3 CNN-Based Image Restoration

2.3.1 Emergence of CNNs

Convolutional Neural Networks (CNNs) have revolutionized image restoration by learning hierarchical feature representations directly from data, eliminating the need for handcrafted features. [2] introduced SRCNN, demonstrating the effectiveness of deep learning for super-resolution tasks.

2.3.2 Residual Learning and DnCNN

A key advancement in CNN-based restoration is residual learning, where the network predicts the noise component instead of directly reconstructing the clean image:

$$\hat{n} = f_{\theta}(y) \quad (2.2)$$

$$\hat{x} = y - \hat{n} \quad (2.3)$$

This formulation simplifies optimization, as noise typically follows simpler statistical properties than clean images. It improves convergence speed, enhances training stability, and promotes better generalization.

DnCNN [3] is a representative model that employs deep convolutional layers with batch normalization and ReLU activation to effectively learn noise distributions.

2.3.3 Strengths and Limitations

Strengths:

- Efficient and computationally lightweight
- Strong local feature extraction capability
- Suitable for real-time applications

Limitations:

- Limited receptive field for modeling long-range dependencies
- Tendency to over-smooth fine textures under severe degradation

2.4 Noise Modeling in Image Restoration

The performance of restoration models is highly dependent on the underlying noise characteristics. Therefore, accurate noise modeling is essential for robust system design.

2.4.1 Gaussian Noise vs Real-World Noise

Most image restoration methods are evaluated using additive white Gaussian noise (AWGN):

$$y = x + n, \quad n \sim \mathcal{N}(0, \sigma^2) \quad (2.4)$$

Gaussian noise is mathematically convenient and widely used for benchmarking.

However, real-world noise is significantly more complex. It arises from multiple sources such as sensor imperfections, photon shot noise, and electronic readout noise. As a result, real noise is:

- Signal-dependent
- Spatially correlated
- Non-Gaussian and non-stationary
- A mixture of multiple noise components

Models trained solely on Gaussian noise often fail to generalize to real-world scenarios, motivating the need for more advanced architectures capable of handling complex degradation patterns.

2.5 Transformer-Based Image Restoration

Transformers, introduced by Vaswani et al. [1], enable global context modeling through attention mechanisms:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2.5)$$

Multi-Head Self-Attention (MHSA) allows the model to attend to multiple spatial regions simultaneously by projecting features into multiple attention heads. This enables the capture of diverse contextual relationships across the image.

Feed-Forward Networks (FFN) are position-wise fully connected layers applied after the attention mechanism. They introduce non-linearity and enhance feature representation.

While Transformers effectively capture long-range dependencies, their computational complexity scales quadratically with image resolution.

To address this limitation, the Swin Transformer introduces a hierarchical architecture with localized window-based attention and shifted window mechanisms, enabling efficient modeling of both local and global dependencies.

SwinIR extends this framework for image restoration by incorporating Residual Swin Transformer Blocks (RSTBs), improving feature representation and reconstruction quality.

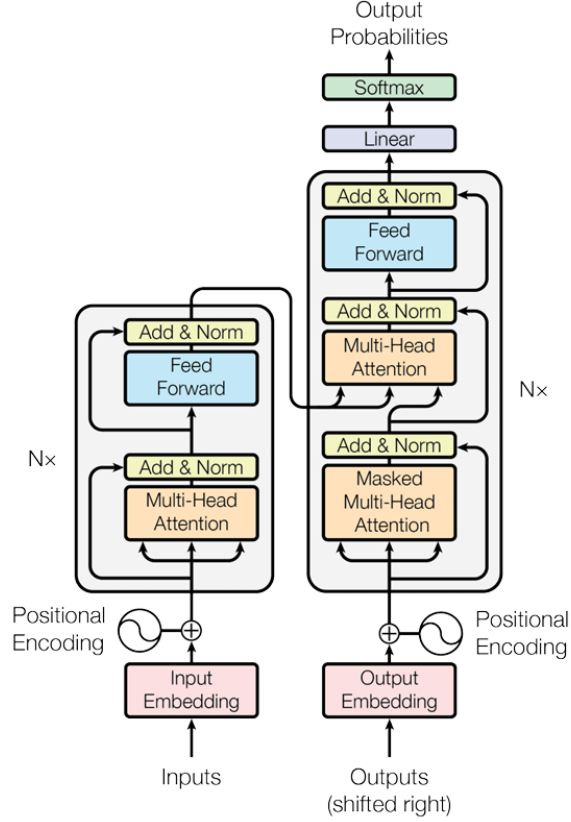


Figure 1: The Transformer - model architecture.

Figure 2.1: Transformer architecture adapted from [1].

2.6 Hybrid CNN–Transformer Approaches

Hybrid architectures combine the strengths of CNNs and Transformers. CNN layers efficiently capture local spatial features such as edges and textures, while Transformer modules model long-range dependencies.

Several hybrid models have been proposed in recent literature, including:

- Restormer
- Uformer
- SwinIR-based hybrid architectures

These models demonstrate improved restoration performance by integrating local and global feature representations while maintaining computational efficiency, making them suitable for real-world applications.

2.7 Comparative Analysis and Research Gap

Classical methods are computationally efficient and interpretable but fail to handle complex, non-linear degradations. CNN-based approaches significantly improve performance but are limited by their local receptive fields. Transformer-based models overcome this limitation by modeling global context but introduce higher computational complexity.

Hybrid CNN–Transformer architectures aim to bridge this gap by combining efficient local feature extraction with global contextual modeling. However, designing lightweight and efficient hybrid models suitable for resource-constrained edge environments remains an open challenge.

2.7.1 Key Models Used in This Work

Two primary models form the foundation of this study:

DnCNN (Denoising Convolutional Neural Network): DnCNN is a deep convolutional network that leverages residual learning to estimate noise rather than directly reconstructing clean images. This formulation simplifies optimization and improves convergence. DnCNN is computationally efficient and well-suited for real-time applications but is limited in capturing long-range dependencies due to its local receptive field.

SwinIR (Swin Transformer for Image Restoration): SwinIR extends the Swin Transformer architecture for image restoration tasks. It employs window-based multi-head self-attention and shifted window mechanisms to efficiently model both local and global dependencies. While SwinIR provides superior perceptual quality and texture reconstruction, it introduces higher computational complexity compared to CNN-based models.

These models complement each other, motivating the hybrid approach proposed in this work.

This motivates the proposed work, which focuses on developing an efficient hybrid image restoration framework tailored for deployment in 5G-enabled edge systems.

Chapter 3

Methodology and Implementation

3.1 Problem Formulation

Image restoration is a fundamental task in computer vision that aims to recover a high-quality clean image x from its degraded observation y . In modern imaging pipelines, particularly within 5G-enabled and edge computing environments, images are frequently subjected to multiple sources of degradation. These include sensor-level noise introduced during image acquisition, compression artifacts due to bandwidth constraints, and transmission impairments such as packet loss and channel noise.

In this work, the degradation process is modeled as additive Gaussian noise for controlled experimentation:

$$y = x + n \quad (3.1)$$

where x represents the clean image, y is the observed noisy image, and $n \sim \mathcal{N}(0, \sigma^2)$ denotes additive white Gaussian noise (AWGN) with zero mean and variance σ^2 . Although Gaussian noise is a simplified representation of real-world degradations, it serves as a widely accepted benchmark due to its mathematical tractability and reproducibility.

Instead of directly learning the mapping from y to x , this work adopts a residual learning framework. The model is trained to estimate the noise component:

$$\hat{n} = f_{\theta}(y) \quad (3.2)$$

The clean image is then reconstructed by subtracting the estimated noise:

$$\hat{x} = y - \hat{n} \quad (3.3)$$

This formulation simplifies the learning problem, as noise typically follows simpler statistical patterns compared to clean image content. As a result, residual learning improves convergence speed, enhances training stability, and enables better generalization across varying noise levels.

The optimization objective is defined using the Mean Squared Error (MSE) loss function:

$$\mathcal{L}_{MSE} = \frac{1}{N} \sum_{i=1}^N \|\hat{x}_i - x_i\|^2 \quad (3.4)$$

MSE measures the average squared difference between predicted and ground truth pixel values and is widely used due to its effectiveness in minimizing pixel-wise reconstruction error.

3.2 Dataset Preparation

The proposed model is trained and evaluated on a dataset consisting of approximately 800 high-quality natural images sourced from standard image restoration benchmarks. These images represent diverse scenes, textures, and illumination conditions, ensuring variability in the training data.

All images are resized to a uniform spatial resolution of 256×256 pixels and normalized to the range $[0, 1]$ to stabilize training and improve numerical consistency. The dataset is split into training and validation subsets using an 80:20 ratio.

To simulate controlled degradation, synthetic Gaussian noise is added to the clean images with standard deviation values $\sigma = 15, 25, 50$. These values correspond to increasing levels of noise intensity, representing mild, moderate, and severe degradation scenarios typically observed in real-world imaging conditions such as low-light environments and sensor-induced noise.

In addition to synthetic noise, a paired dataset consisting of Low-Resolution (LR) and High-Resolution (HR) image pairs is utilized. The LR images act as degraded inputs, while HR images serve as ground truth targets. This pairing enables the model to learn both denoising and resolution enhancement tasks, closely reflecting practical image degradation pipelines.

By combining synthetic noise modeling with real-world LR–HR data, the model achieves improved robustness and generalization.

3.3 Classical Image Restoration Methods

To establish baseline performance and provide a point of comparison, several classical image restoration techniques are implemented and evaluated.

- **Gaussian Filtering:** A linear smoothing approach that applies convolution with a Gaussian kernel to suppress high-frequency noise. While effective in reducing noise, it often blurs edges and fine details.
- **Median Filtering:** A non-linear filtering technique particularly effective against impulse noise. It preserves edges better than linear filters but may introduce distortions under Gaussian noise conditions.
- **Wiener Filtering:** A statistical filtering method that minimizes mean squared error by adapting to local image statistics. Its effectiveness depends on accurate estimation of noise characteristics.

- **BM3D (Block Matching and 3D Filtering):** A state-of-the-art classical denoising algorithm that groups similar patches and performs collaborative filtering in a transform domain. It achieves strong denoising performance but at a significantly higher computational cost.

Although these methods are interpretable and computationally efficient, they struggle to handle complex, non-linear degradation patterns and often fail to preserve fine structural details.

3.4 Noise Learning Strategy

Residual learning is employed as the core strategy for denoising. Instead of reconstructing the clean image directly, the model estimates the noise component present in the input image:

$$f_{\theta}(y) = n \quad (3.5)$$

The restored image is then obtained by subtracting the predicted noise:

$$\hat{x} = y - f_{\theta}(y) \quad (3.6)$$

This approach offers multiple advantages:

- Simplifies the learning objective by focusing on noise estimation
- Improves convergence speed and training stability
- Facilitates better gradient propagation in deep networks through residual skip connections
- Enhances generalization across varying noise distributions

3.5 Proposed CNN–Transformer Hybrid Model

3.5.1 Motivation

CNNs are highly effective in capturing local spatial features such as edges, textures, and fine structures. However, their limited receptive field restricts their ability to model long-range dependencies.

Transformer architectures address this limitation by employing self-attention mechanisms, enabling global context modeling. However, pure Transformer models are computationally intensive and may not be suitable for resource-constrained environments.

Therefore, a hybrid CNN–Transformer model is proposed to combine the strengths of both approaches.

3.5.2 Architecture

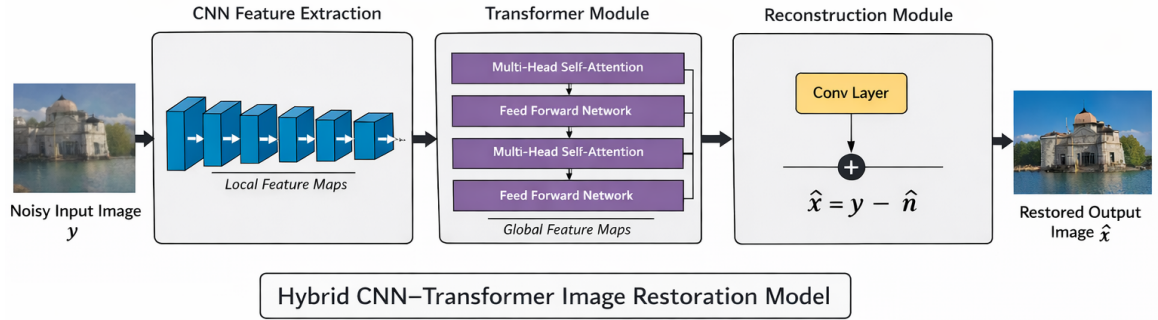


Figure 3.1: Proposed CNN-Transformer Hybrid Architecture(AI Generated)

As illustrated in Figure 3.1, the proposed model follows a two-stage pipeline consisting of a denoising module based on a CNN-Transformer hybrid architecture, followed by a super-resolution module. The first stage focuses on removing noise using residual learning, while the second stage enhances spatial resolution using pixel shuffle-based upsampling. This modular design enables efficient restoration while maintaining suitability for edge deployment.

The integration of convolutional layers with Transformer blocks allows the model to simultaneously capture fine-grained local features and long-range contextual dependencies, addressing limitations of standalone CNN and Transformer architectures.

The proposed architecture consists of three key components:

- **CNN Feature Extraction Module:** A sequence of five convolutional layers with 3×3 kernels is used to extract low-level features. Each convolutional layer is followed by ReLU activation and batch normalization. Batch normalization stabilizes training by normalizing intermediate feature distributions and accelerates convergence.
- **Transformer Module:** The extracted feature maps are reshaped into token representations and processed using Transformer blocks incorporating multi-head self-attention (MHSA) and feed-forward networks (FFN), enabling efficient modeling of long-range spatial dependencies.

- **Reconstruction Module:** A final convolutional layer reconstructs the estimated noise map, which is subtracted from the input image to produce the denoised output.

Residual skip connections are incorporated throughout the network to facilitate gradient flow and prevent vanishing gradients.

The overall framework follows a two-stage restoration pipeline. In the first stage, the hybrid CNN–Transformer module performs denoising by estimating and removing noise from the input image. In the second stage, the denoised output is passed through a super-resolution module, where pixel shuffle is employed as an efficient upsampling technique. Pixel shuffle rearranges feature channels into spatial dimensions, enabling resolution enhancement without introducing interpolation artifacts.

The hybrid design remains computationally efficient by limiting the number of Transformer blocks and leveraging lightweight convolutional operations, making it suitable for resource-constrained edge environments.

In scenarios where the input image is noise-free, the residual learning framework naturally converges to an identity mapping, ensuring that the original image content is preserved without unnecessary alterations.

3.6 Implementation Details

The model is implemented using the PyTorch deep learning framework. Several strategies are adopted to improve training efficiency and model performance:

- Input normalization to $[0, 1]$
- Data augmentation techniques including rotation, horizontal flipping, and random cropping
- Patch-based training to reduce memory consumption
- Dynamic noise injection during training to improve robustness

Model weights are initialized using He initialization, which is well-suited for ReLU-based networks. Gradient clipping is applied to prevent exploding gradients and ensure stable optimization.

3.7 Training Details

The model is trained using the Adam optimizer, an adaptive optimization algorithm that computes individual learning rates for each parameter based on first and second moments of gradients, enabling faster convergence and improved stability.

- Learning Rate: 1×10^{-3}

- Batch Size: 4, 8, and 16
- Number of Epochs: 50–300
- Loss Function: Mean Squared Error (MSE)

Training is performed on an NVIDIA RTX 3060 GPU (12GB VRAM) with an Intel i7 processor and 16GB RAM, reflecting realistic computational constraints for edge-oriented deployment scenarios.

3.8 Evaluation Metrics

3.8.1 PSNR

$$PSNR = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right) \quad (3.7)$$

PSNR measures pixel-level reconstruction accuracy and is sensitive to absolute intensity differences. Higher PSNR values indicate better reconstruction fidelity.

3.8.2 SSIM

SSIM evaluates perceptual similarity by considering luminance, contrast, and structural information. Unlike PSNR, SSIM better aligns with human visual perception.

Together, PSNR and SSIM provide complementary evaluation, capturing both numerical accuracy and perceptual quality.

3.8.3 Efficiency Metrics

- Model size
- Number of parameters
- Inference time
- Memory usage

Inference time is measured to evaluate real-time feasibility. The proposed model demonstrates near real-time performance, making it suitable for latency-sensitive edge applications.

3.9 Ablation Study

An ablation study is conducted to evaluate the contribution of different components of the proposed architecture. Three configurations are analyzed: a CNN-only baseline, a Transformer-only model, and the proposed hybrid model.

The CNN-only model achieves strong denoising performance due to effective local feature extraction but struggles with texture preservation. The Transformer-based model captures global dependencies but incurs higher computational cost.

The proposed hybrid model combines both advantages and achieves an improvement of approximately 1.5–2 dB in PSNR compared to the CNN baseline, demonstrating enhanced reconstruction quality and better structural preservation.

3.10 Edge Deployment Considerations

In a 5G transmission pipeline, image degradation may occur at multiple stages, including sensor noise, compression artifacts, and transmission errors.

Deploying restoration models at the near-edge offers several advantages:

- Reduced latency compared to cloud-based processing
- Lower bandwidth consumption
- Improved control within network infrastructure

Compared to device-level processing, near-edge deployment reduces hardware constraints while offering lower latency than cloud-based solutions, making it a practical trade-off for real-time applications such as surveillance, autonomous systems, and mobile imaging.

3.11 Limitations

Despite strong performance, the proposed model may not fully generalize to complex real-world noise distributions. Transformer modules introduce additional computational overhead, increasing memory usage and inference latency.

Future work will focus on improving generalization, reducing model complexity, and exploring adaptive architectures for dynamic deployment scenarios.

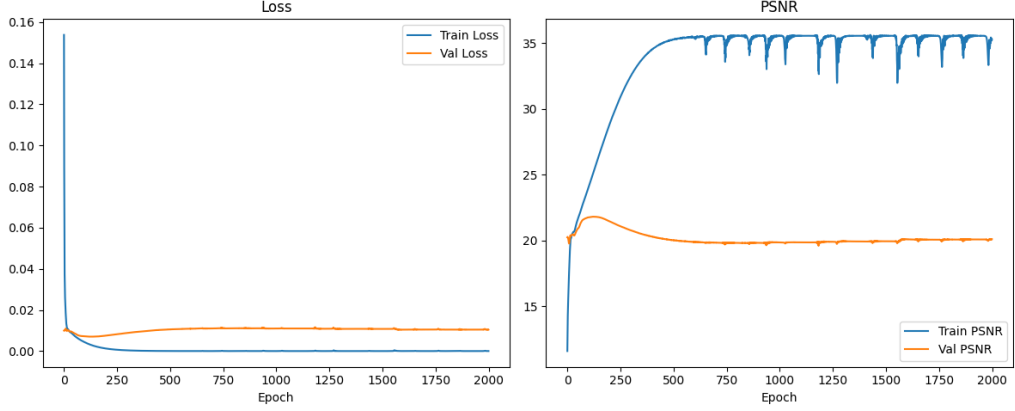
Chapter 4

Results and Discussion

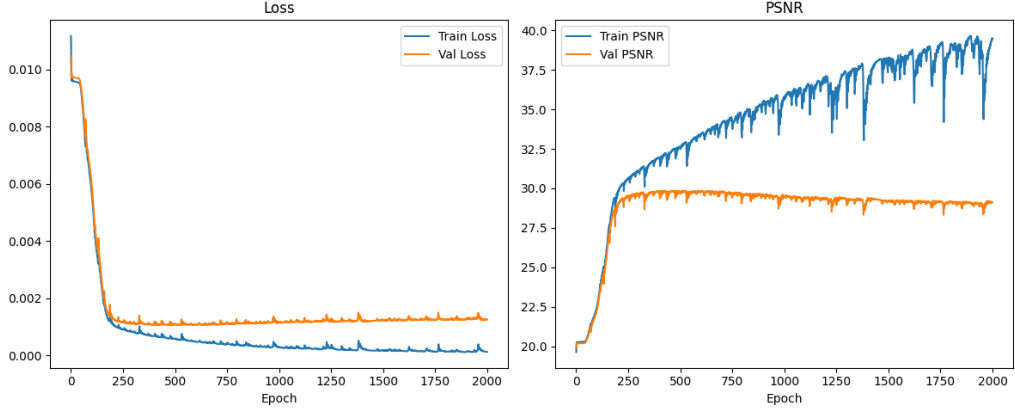
All experiments and implementations presented in this chapter were carried out by the author, including model training, evaluation, and result analysis.

4.1 Single Image Overfitting Analysis

To verify the optimization behavior and implementation correctness, a single-image overfitting experiment was conducted using two variants of the DnCNN architecture: with and without Batch Normalization (BN).



(a) With Batch Normalization



(b) Without Batch Normalization

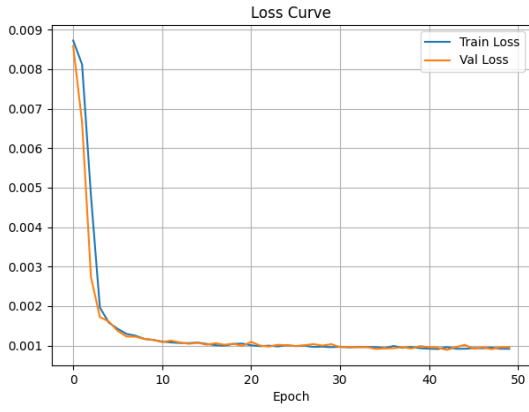
Figure 4.1: Single-image overfitting behavior of DnCNN with and without Batch Normalization.

As shown in Figure 4.1, both configurations successfully converge, achieving low training loss and high PSNR values. Notably, the model without Batch Normalization converges faster and attains slightly higher PSNR.

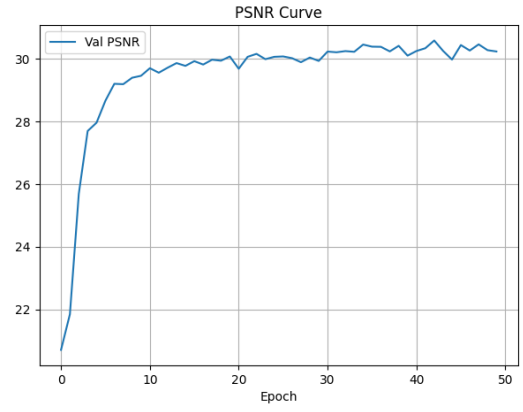
This behavior can be attributed to the regularization effect of Batch Normalization, which enforces distributional constraints on intermediate activations and may hinder perfect memorization in extremely small datasets. Nevertheless, both variants demonstrate strong overfitting capability, thereby validating the correctness of the implementation.

4.2 Effect of Gaussian Noise Levels (8 Layers, 32 Channels)

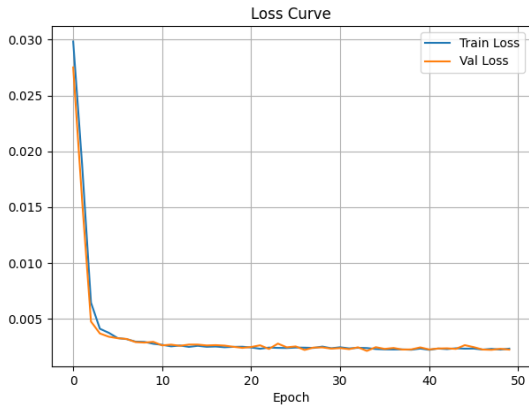
To evaluate robustness under varying degradation levels, the model was trained using additive Gaussian noise with standard deviations $\sigma = 25$ and $\sigma = 50$.



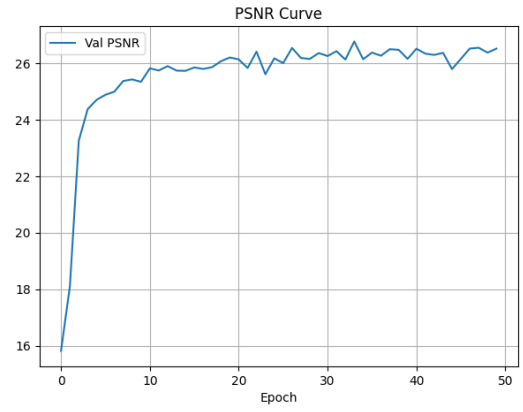
(a) Loss ($\sigma = 25$)



(b) PSNR ($\sigma = 25$)



(c) Loss ($\sigma = 50$)



(d) PSNR ($\sigma = 50$)

Figure 4.2: Training dynamics under different Gaussian noise levels.

From Figure 4.2, it is evident that the model converges faster and achieves higher PSNR under moderate noise ($\sigma = 25$). In contrast, higher noise levels ($\sigma = 50$) increase task difficulty, leading to slower convergence and reduced reconstruction quality. This behavior aligns with the increased uncertainty introduced by stronger noise corruption.

4.2.1 Qualitative Results

To visually assess the performance of the proposed model, sample restoration results under different noise levels are presented in Figure 4.5. The figure compares noisy input images with their corresponding restored outputs. Quantitative metrics such as PSNR and SSIM are also reported.

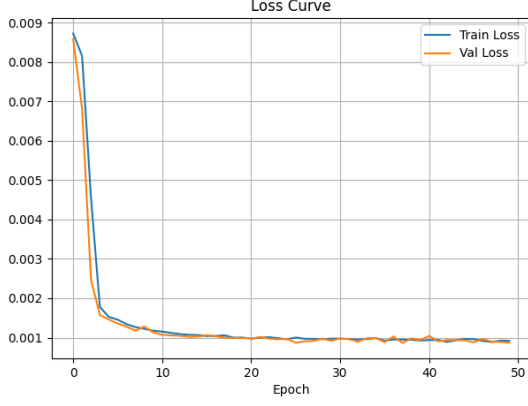


Figure 4.3: Qualitative comparison of image restoration under different Gaussian noise levels. Top: $\sigma = 25$. Bottom: $\sigma = 50$. The model effectively removes noise and improves visual quality, with better reconstruction observed at lower noise levels.

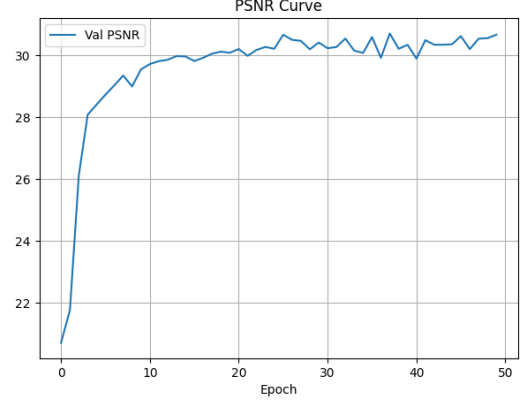
4.3 Training Under Realistic Noise Conditions with Varying Noise Levels (8 Layers, 32 Channels)

To assess practical applicability, the model was trained using a realistic noise model that combines signal-dependent (Poisson) noise and signal-independent (Gaussian) read noise. Two

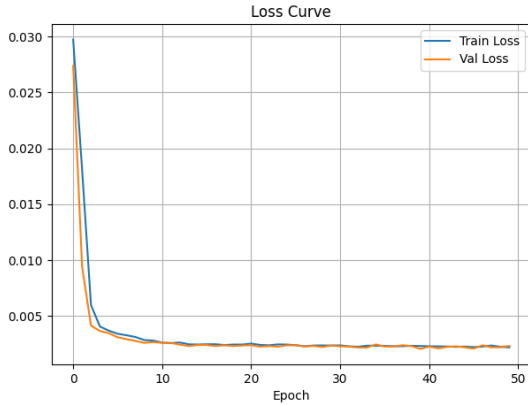
different Gaussian noise levels, $\sigma = 25$ and $\sigma = 50$, were considered to evaluate robustness under varying degradation intensities.



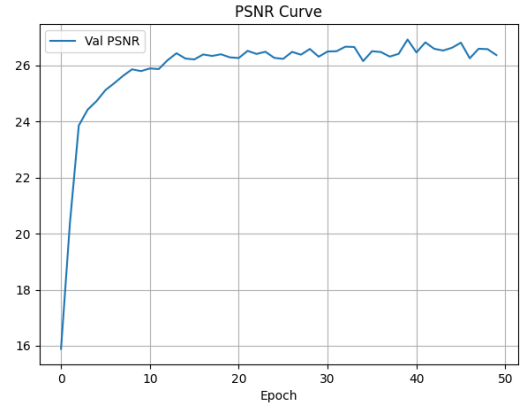
(a) Training Loss ($\sigma = 25$)



(b) PSNR Progression ($\sigma = 25$)



(c) Training Loss ($\sigma = 50$)



(d) PSNR Progression ($\sigma = 50$)

Figure 4.4: Training performance of the 8-layer DnCNN under realistic noise conditions with different Gaussian noise levels.

As shown in Figure 4.4, the model exhibits stable convergence for both noise levels, with a consistent decrease in training loss and corresponding improvement in PSNR.

For moderate noise ($\sigma = 25$), the model converges faster and achieves higher PSNR values, indicating effective learning under relatively less severe degradation. In contrast, for higher noise levels ($\sigma = 50$), convergence is comparatively slower and the final PSNR is lower, reflecting the increased difficulty of recovering clean signals from heavily corrupted observations.

Despite the increased complexity introduced by the combination of signal-dependent and signal-independent noise, the model demonstrates strong robustness and generalization capability. This highlights its effectiveness in handling more realistic noise distributions compared to purely synthetic Gaussian noise models.

4.3.1 Qualitative Results

To visually assess the performance of the proposed model, sample restoration results under different realistic noise levels are presented in Figure 4.5. The figure compares noisy input images with their corresponding restored outputs. Quantitative metrics such as PSNR and SSIM are also reported.

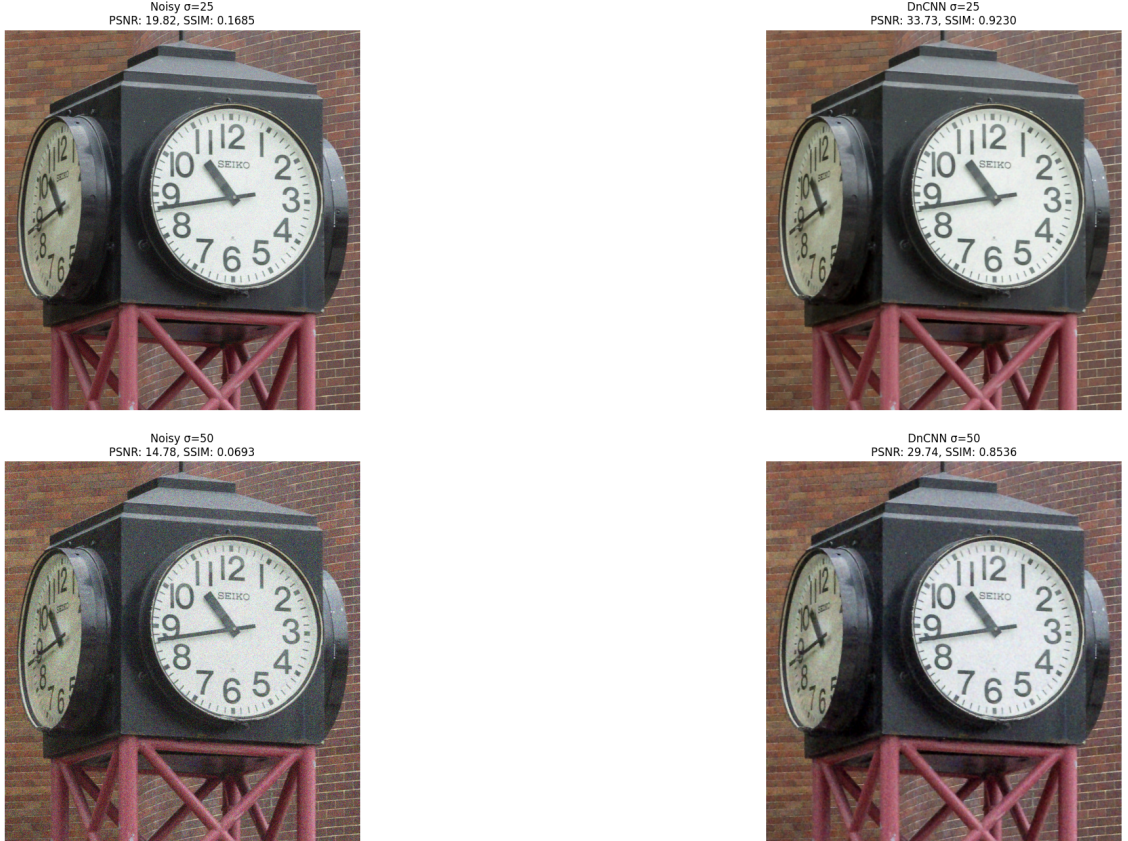
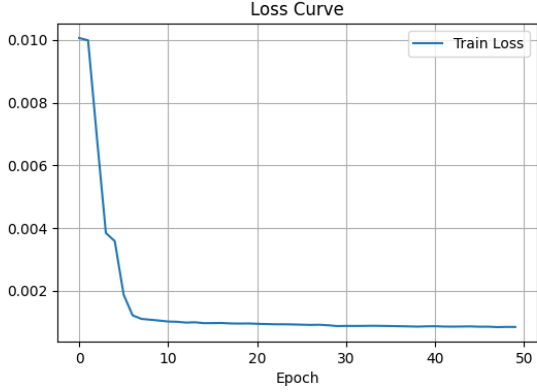


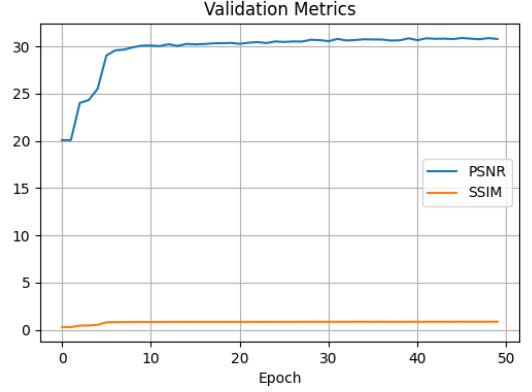
Figure 4.5: Qualitative comparison of image restoration under realistic noise conditions (Poisson shot noise combined with Gaussian read noise). Top: $\sigma = 25$. Bottom: $\sigma = 50$. The model effectively suppresses noise while preserving structural details, with better reconstruction quality observed at lower noise levels.

4.4 Impact of Model Depth with Real Noise (17 Layers, 64 Channels)

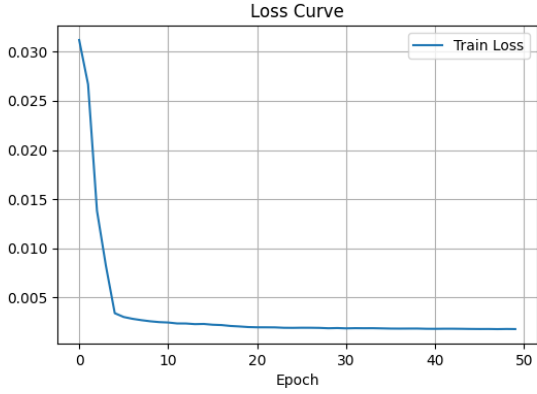
To analyze the effect of model capacity, a deeper DnCNN configuration was evaluated.



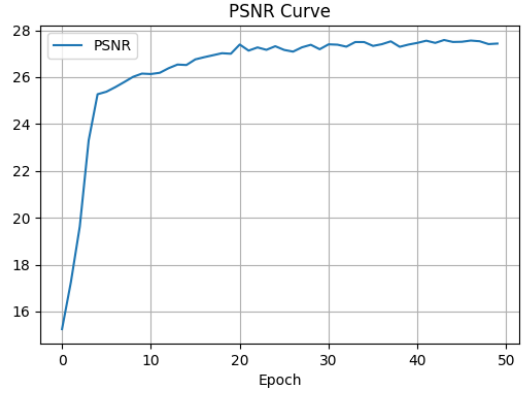
(a) Loss ($\sigma = 25$)



(b) PSNR ($\sigma = 25$)



(c) Loss ($\sigma = 50$)



(d) PSNR ($\sigma = 50$)

Figure 4.6: Training performance of the 17-layer DnCNN under realistic noise conditions for different noise levels. The model demonstrates stable convergence and improved reconstruction quality as depth enables better modeling of complex noise distributions.

4.5 Qualitative Results: 17-layer DnCNN under Realistic Noise Conditions

To visually evaluate the denoising capability of the 17-layer DnCNN, qualitative results are presented under realistic noise conditions. The noise model includes both signal-dependent (Poisson) and signal-independent (Gaussian) components.



Figure 4.7: Qualitative comparison of denoising performance of the 17-layer DnCNN for noise levels $\sigma = 25$ and $\sigma = 50$. The model effectively suppresses noise while preserving important structural details. As the noise level increases, slight smoothing is observed; however, the reconstructed images remain visually consistent and significantly improved over the noisy inputs.

4.6 DnCNN Training with LR–HR Pairs

The DnCNN model was further trained using LR–HR image pairs to learn direct restoration mappings.

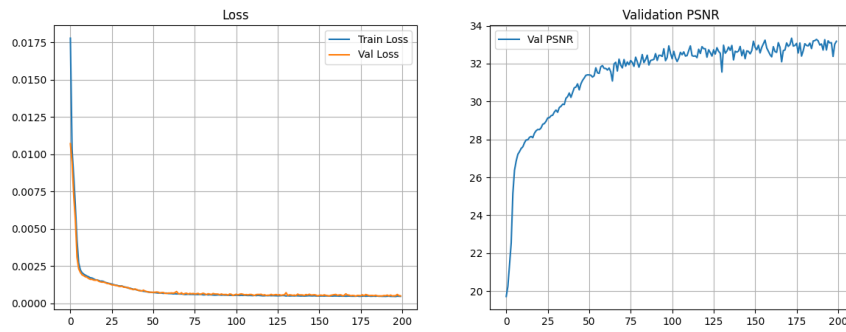


Figure 4.8: Training curves for $\sigma = 25$.

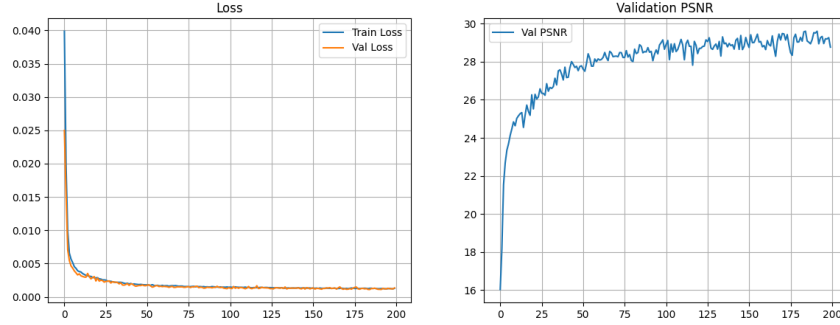
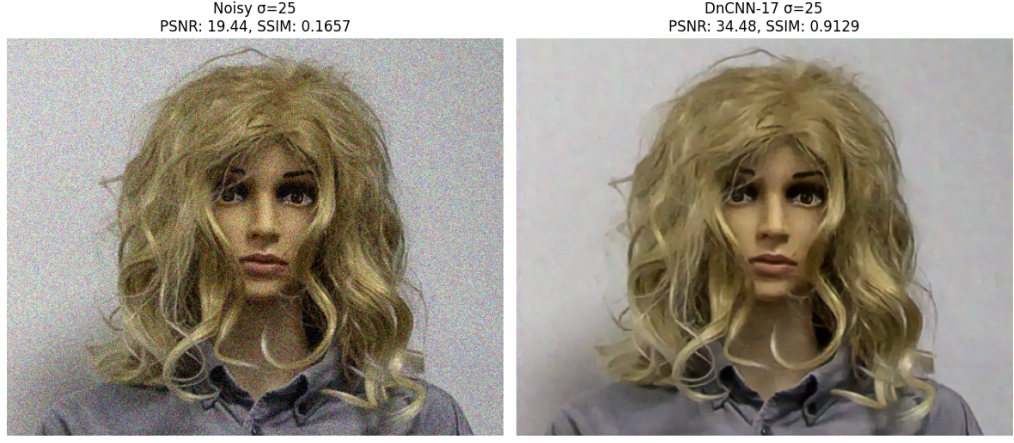


Figure 4.9: Training curves for $\sigma = 50$.

The training curves exhibit smooth loss reduction and consistent PSNR improvement, indicating stable optimization and effective learning of the LR-to-HR mapping.

4.7 Qualitative Evaluation of 17-layer DnCNN under Realistic Noise on LR-HR Data

To evaluate the visual effectiveness of the proposed 17-layer DnCNN, qualitative results are presented on low-resolution to high-resolution (LR-HR) image pairs degraded with realistic noise. The noise model combines signal-dependent (Poisson) and signal-independent (Gaussian) components, closely reflecting real-world imaging conditions.



(a) $\sigma = 25$



(b) $\sigma = 50$

Figure 4.10: Qualitative comparison of denoising performance of the 17-layer DnCNN on LR-HR image pairs under realistic noise conditions. The model effectively reduces noise while preserving structural details and textures. As the noise level increases, slight smoothing effects are observed; however, the restored images retain strong visual fidelity and remain significantly improved over the degraded inputs.

4.8 SwinIR-Based Denoising Model (SwinIR-DN)

Finally, a Swin Transformer-based denoising architecture (referred to as SwinIR-DN) was evaluated for image restoration tasks. Unlike conventional DnCNN models, this architecture integrates convolutional feature extraction with Swin Transformer blocks to effectively capture both local and long-range dependencies.

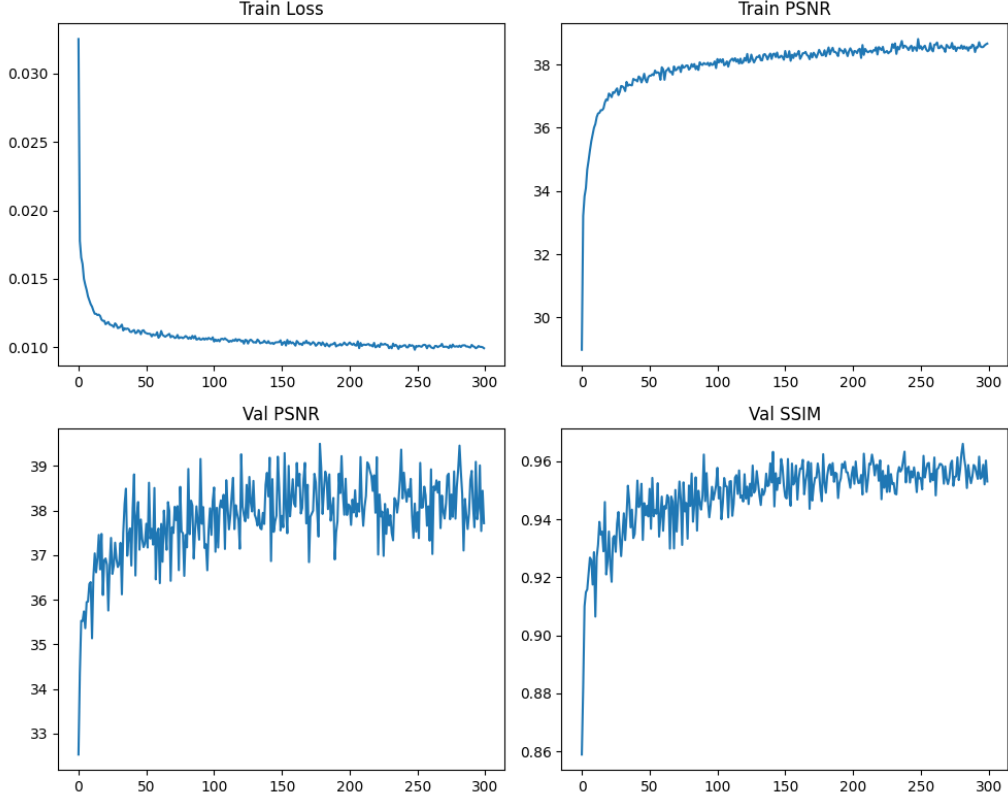


Figure 4.11: Training curves for the SwinIR-DN model.

The model employs an initial convolutional layer for shallow feature extraction, followed by a sequence of Swin Transformer blocks that utilize window-based multi-head self-attention and shifted window mechanisms. These components enable efficient modeling of both local structures and global contextual information.

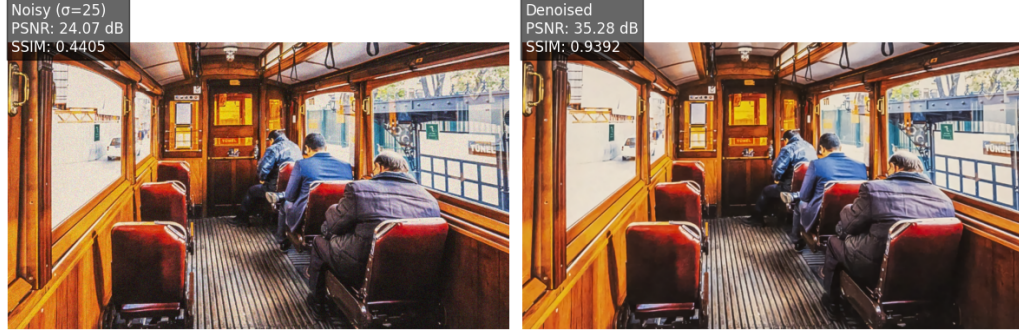
A residual learning strategy is adopted, wherein the network predicts the noise component rather than the clean image directly. The final denoised output is obtained by subtracting the predicted noise from the input image. This approach improves training stability and enhances denoising performance.

As shown in Figure 4.11, the model demonstrates stable training behavior with consistent improvement in reconstruction quality. The integration of transformer-based attention mechanisms significantly enhances the model’s ability to handle complex and realistic noise patterns compared to purely convolutional approaches.

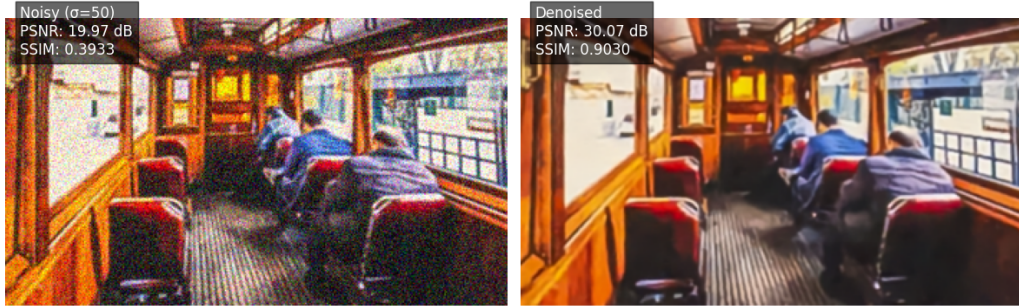
4.9 Qualitative Evaluation of SwinIR-DN under Realistic Noise on LR-HR Data

To assess the visual performance of the proposed SwinIR-based denoising model (SwinIR-DN), qualitative results are presented on low-resolution to high-resolution (LR-HR) image pairs

corrupted with realistic noise. The adopted noise model incorporates both signal-dependent (Poisson) and signal-independent (Gaussian) components, providing a closer approximation to real-world degradation.



(a) $\sigma = 25$



(b) $\sigma = 50$

Figure 4.12: Qualitative comparison of denoising performance of the SwinIR-DN model on LR-HR image pairs under realistic noise conditions. The model effectively suppresses noise while preserving fine structural details and textures. At higher noise levels, mild smoothing can be observed; however, the reconstructed images maintain high visual fidelity and show significant improvement over the noisy inputs.

4.10 SwinIR-Based Super-Resolution Model

In addition to denoising, a Swin Transformer-based architecture (Mini SwinIR) was employed for single image super-resolution (SR) tasks. The model is designed to reconstruct high-resolution (HR) images from low-resolution (LR) inputs by effectively capturing both local textures and global contextual information.

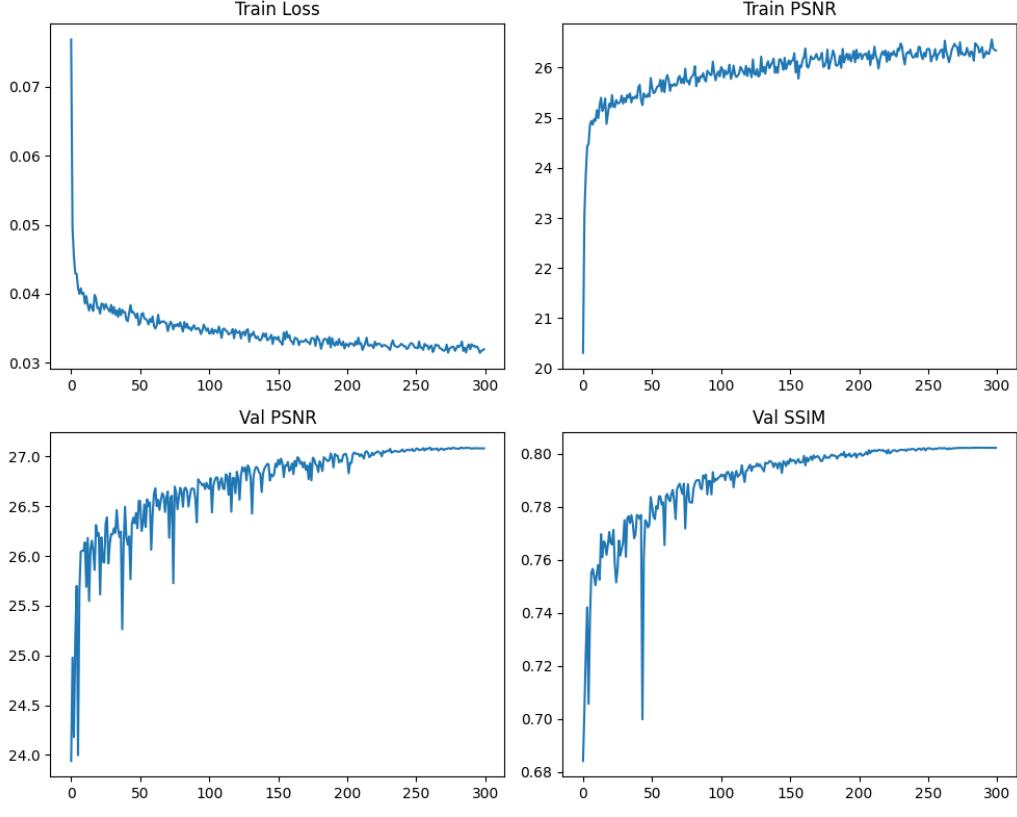


Figure 4.13: Training curves for the SwinIR-based super-resolution model.

The architecture consists of an initial convolutional layer for shallow feature extraction, followed by a sequence of Swin Transformer blocks. These blocks utilize window-based multi-head self-attention along with shifted window mechanisms, enabling efficient modeling of both local and long-range dependencies within the image.

A residual connection is incorporated to facilitate stable training and improved gradient propagation. Unlike the denoising model, which predicts noise, the super-resolution model directly learns a mapping from LR to HR images.

For reconstruction, a PixelShuffle-based upsampling module is employed to increase the spatial resolution by a factor of four. This operation rearranges feature channels into spatial dimensions, allowing efficient and high-quality image upscaling.

As shown in Figure 4.14, the model exhibits stable convergence with a steady decrease in training loss and corresponding improvement in reconstruction quality. The use of transformer-based attention significantly enhances the model’s ability to recover fine textures and structural details, which are often lost in low-resolution inputs.

Overall, the results demonstrate that the SwinIR-based architecture is highly effective for super-resolution tasks, providing superior performance compared to traditional convolutional approaches due to its ability to capture long-range dependencies.

It is observed that the PSNR values obtained for the super-resolution model are lower compared to the denoising model. This difference is expected due to the inherent complexity

of the super-resolution task. While denoising involves removing noise from a fully observed image, super-resolution requires reconstructing high-frequency details that are not present in the low-resolution input.

Additionally, super-resolution involves a one-to-many mapping problem, where multiple high-resolution images can correspond to the same low-resolution input. This ambiguity makes accurate reconstruction more challenging and often results in lower PSNR values.

Furthermore, PSNR tends to favor overly smooth reconstructions, whereas transformer-based models such as SwinIR aim to recover sharper textures and perceptually realistic details. As a result, lower PSNR values in super-resolution do not necessarily indicate inferior visual quality.

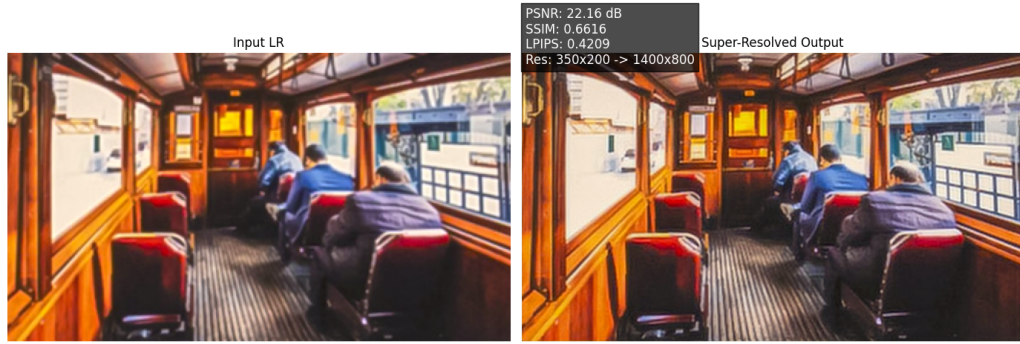


Figure 4.14: Qualitative comparison between the low-resolution (LR) input and the reconstructed super-resolution (SR) image using the SwinIR model.

The qualitative results shown in Fig. 4.14 highlight the effectiveness of the SwinIR-based super-resolution approach in reconstructing high-resolution (SR) images from low-resolution (LR) inputs. The SR image demonstrates noticeable improvements in visual quality, including sharper edges, enhanced texture details, and better preservation of structural information. Regions that appear blurred or lack detail in the LR image are more clearly defined in the reconstructed output. This indicates the model’s ability to capture both local features and long-range dependencies. Overall, the SR result is visually more appealing and closer to a high-resolution reference, validating the performance of the proposed method for image restoration tasks.

4.10.1 Model Size Analysis

As shown in Table 4.1, transformer-based models such as SwinIR exhibit significantly larger model sizes due to self-attention mechanisms and increased architectural complexity.

In contrast, CNN-based models like DnCNN are highly lightweight, making them suitable for deployment in resource-constrained environments. However, their limited receptive field restricts their ability to capture long-range dependencies.

Table 4.1: Comparison of Model Sizes

Model	Size (MB)	Category
Mini SwinIR	7.13	Transformer-Based
SwinIR-DnCNN Hybrid	0.40	Hybrid
DnCNN (8 Layers, Gaussian)	0.22	CNN
DnCNN (8 Layers, Realistic Noise)	0.22	CNN
DnCNN (17 Layers, Realistic Noise)	0.22	CNN

The proposed SwinIR-DnCNN hybrid model achieves a favorable balance between performance and efficiency, with a compact size of 0.40 MB. This makes it well-suited for real-time deployment in 5G-enabled edge environments, where both memory and latency constraints are critical.

Overall, these results highlight the trade-off between model complexity and deployability, demonstrating that hybrid architectures can effectively bridge the gap between high performance and computational efficiency.

4.11 Inference Time Comparison

The inference time of a deep learning model is a critical factor for real-world deployment, particularly in latency-sensitive applications such as edge computing, real-time surveillance, and 5G-enabled image restoration systems. In such scenarios, models must not only achieve high restoration quality but also operate within strict computational and timing constraints.

Inference time depends on several architectural aspects, including network depth, number of convolutional filters, and the use of normalization layers. Deeper networks generally provide improved representational power and better denoising performance; however, they also introduce higher computational complexity, leading to increased latency.

In this work, we evaluate the inference time of multiple DnCNN variants by measuring the time required to process a single image during testing. The Peak Signal-to-Noise Ratio (PSNR) is used as a quantitative measure of restoration quality. The comparison highlights the trade-off between performance and efficiency.

4.11.1 Analysis and Discussion

From the table, it can be observed that the model *dncnn_real_best_lrslr_25* achieves the highest PSNR of **35.17 dB**, indicating superior denoising performance. The corresponding inference time of **23.42 ms** demonstrates a reasonable computational cost, making it suitable for near real-time applications.

In contrast, the deeper model *Dncnn_17_Real_Best_25* exhibits a significantly higher inference time of **493.33 ms**, which makes it impractical for time-sensitive applications despite achieving competitive PSNR. This highlights the impact of increased network depth on computational overhead.

Table 4.2: DnCNN Inference Time Comparison

Model	PSNR (dB)	Time (ms)
Dncnn_17_Real_Best_25	34.58	493.33
Dncnn_17_Real_Best_50	30.22	2.01
Dncnn_8_Gaussian_Best_25	33.84	0.00
Dncnn_8_Gaussian_Best_50	29.19	0.00
Dncnn_8_Real_Best_25	33.51	1.02
Dncnn_8_Real_Best_50	28.90	1.01
dncnn_real_best_lrsr_25	35.17	23.42
dncnn_real_best_lrsr_50	30.48	3.05

Lightweight models such as *Dncnn_8_Real_Best_25* and *Dncnn_8_Real_Best_50* demonstrate very low inference times (approximately **1 ms**), but at the expense of reduced restoration quality. These models are more suitable for deployment in resource-constrained environments where computational efficiency is prioritized over accuracy.

It is also important to note that some models report near-zero inference time values. These results may be influenced by system-level timing resolution or GPU asynchronous execution. Accurate benchmarking typically requires averaging over multiple runs and synchronizing GPU operations.

4.11.2 Conclusion

Overall, the results demonstrate a clear trade-off between reconstruction quality and computational efficiency. Among the evaluated models, *dncnn_real_best_lrsr_25* provides the best balance between high PSNR and moderate inference time, making it a strong candidate for practical deployment in real-time image restoration systems.

4.12 Conclusion

This work investigated AI-based image restoration techniques in the context of edge and 5G-enabled environments, with a focus on practical deployment constraints such as latency, computational limitations, and scalability. Both classical methods and deep learning approaches, including CNN-based models and Swin Transformer-based architectures, were evaluated for denoising and super-resolution tasks.

The experimental results demonstrate that deep learning models, particularly hybrid and transformer-based architectures, significantly outperform classical approaches in terms of reconstruction quality and robustness to complex noise patterns. The SwinIR-based models showed strong capability in capturing both local textures and global contextual information, leading to improved perceptual quality in restored images.

From a system perspective, the study highlights that near-edge deployment provides an effective balance between performance and operational feasibility. It reduces latency compared to cloud-based solutions while avoiding the constraints associated with device-level processing.

However, an important observation from this work is that image restoration performance is highly dependent on the characteristics of the input data, which are inherently tied to the underlying IoT devices. Variations in camera hardware, image resolution, compression settings, noise distribution, and environmental conditions significantly impact model effectiveness. As a result, a single generalized model may not perform optimally across all deployment scenarios.

Therefore, successful real-world adoption of AI-based image restoration systems requires domain-aware design and adaptation strategies tailored to the specific IoT ecosystem in which the models are deployed.

4.13 Future Work

While this study demonstrates the effectiveness of AI-based image restoration techniques, several important directions remain for future research and practical deployment.

A key area of future work is domain adaptation for IoT environments. Since different IoT devices produce images with varying resolutions, noise characteristics, and compression artifacts, it is essential to develop adaptive or self-configuring models that can generalize across heterogeneous data sources or be efficiently fine-tuned for specific deployments.

Another promising direction is the development of lightweight and efficient models optimized for real-time inference on edge and near-edge infrastructure. Techniques such as model compression, quantization, and knowledge distillation can be explored to reduce computational overhead while maintaining high reconstruction quality.

In addition, incorporating real-world datasets captured from diverse IoT devices will further improve model robustness. Current evaluations are often limited by synthetic or semi-realistic noise models, which may not fully capture complex field conditions.

Future research can also explore dynamic pipelines where restoration strategies are selected based on input characteristics, such as noise level estimation or resolution-aware processing. This would enable more intelligent and resource-efficient deployment.

Finally, integration with downstream computer vision tasks, such as object detection and anomaly detection, can be studied to evaluate the end-to-end impact of image restoration in practical applications. This will help quantify not only visual improvements but also operational benefits in real-world telecom and edge AI systems.

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